



Time Series Analysis

Seminar 7: Cointegration and VECM



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Acronyms used in this material

- ▣ **ADF**: Augmented Dickey–Fuller (test)
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ **ECM**: Error Correction Model
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **GDP**: Gross Domestic Product
- ▣ **HUF**: Hungarian forint (ISO 4217 code)
- ▣ **OLS**: Ordinary Least Squares
- ▣ **PLN**: Polish zloty (ISO 4217 code)
- ▣ **ROBOR**: Romanian Interbank Offer Rate
- ▣ **SE**: Standard Error
- ▣ **US**: United States
- ▣ **VAR**: Vector AutoRegression
- ▣ **VECM**: Vector Error Correction Model

Today's question and route

- **Question:** are two or more $I(1)$ series tied by a long-run equilibrium, how fast do they return to it, and who does the adjusting?
 - ▶ this seminar comes **before** Lecture 7: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: Engle–Granger statistics by hand; an ECM and its half-life; a bivariate VECM; Johansen statistics; from a VAR to a VECM
 - ▶ Part B: US interest rates; Romanian consumption and GDP; EUR/RON, EUR/HUF and EUR/PLN; ROBOR, the Romanian 10-year yield and Euribor
 - ▶ Part C: pairs trading on two Romanian banks, and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	an Engle–Granger statistic with two and with three variables	Solved, Proposed	A1
A3, A4	an ECM and a bivariate VECM: speed and half-life	Solved, Proposed	A3
A5, A6	Johansen statistics from eigenvalues; a VAR(2) written as a VECM	Solved, Proposed	A5
B1, B2	Engle–Granger and ECM: US yields; Romanian consumption and GDP	Solved, Proposed	B1
B3, B4	Johansen and VECM: three currencies; three interest rates	Solved, Proposed	B3
C1, C2	pairs trading out of sample; what is wrong in an AI answer?	Proposed	B1, B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
US Treasury yields, 1 and 10 years	FRED (GS1, GS10)	monthly	1960–2026
Household consumption and GDP, Romania	Eurostat (namq_10_gdp)	quarterly, SA	1995–2026
EUR/RON, EUR/HUF, EUR/PLN	BNR reference rates	month-end	2005–2026
ROBOR 3M, Euribor 3M	Eurostat (irt_st_m)	monthly	2010–2026
Romanian 10-year yield	EODHD	daily, monthly mean	2010–2026
Banca Transilvania, BRD	EODHD	daily, adjusted close	2014–2026

- ▣ Levels as 100 ln (consumption, GDP, exchange rates, share prices); interest rates in % per year
- ▣ In the notebook: `adf_test`, `eg_test` (Engle–Granger and Phillips–Ouliaris), `ecm_fit`, `johansen`; no account or key is needed

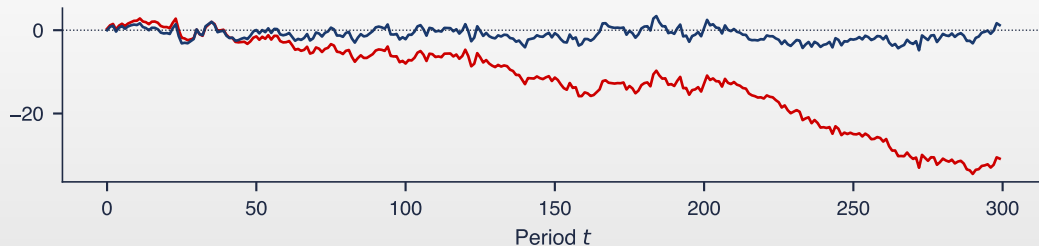
Prerequisites for Today (1/15): unit roots and the ADF test

- $y_t \sim I(1)$: y_t has a unit root and its difference $\Delta y_t = y_t - y_{t-1}$ is stationary, $I(0)$ (Chapter 3)
 - ▶ $I(d)$: stationary after d differences; a shock to an $I(1)$ series never dies out
- **ADF test** (augmented Dickey–Fuller): a regression of the change on the lagged level

$$\Delta y_t = c [+ \delta t] + \gamma y_{t-1} + \sum_{j=1}^k \delta_j \Delta y_{t-j} + e_t, \quad \tau = \frac{\hat{\gamma}}{\text{SE}(\hat{\gamma})}$$

- ▶ c : a constant; δt : an optional linear trend; δ_j : the coefficients of the k lagged differences, which make e_t white noise
- ▶ $\gamma = \phi - 1$ for an AR(1) $y_t = \phi y_{t-1} + \dots$; H_0 : $\gamma = 0$ (unit root); H_1 : $\gamma < 0$ (stationary)
- ▶ SE: the standard error; reject H_0 if τ is below the Dickey–Fuller critical value (constant, 5%: -2.86), not -1.645

Prerequisites for Today (2/15): a random walk and a stationary AR(1)



— Random walk $y_t = y_{t-1} + \varepsilon_t, I(1)$ — AR(1) $y_t = 0.9 y_{t-1} + \varepsilon_t, I(0)$

- The same shocks ε_t ; only the coefficient of y_{t-1} differs: 1 or 0.9
- AR(1): returns to its mean 0, stays within about ± 5
 - ▶ random walk: wanders away (down to -35) and has no mean to return to; its variance grows as $t\sigma^2$

Prerequisites for Today (3/15): the spurious regression

- ▣ **Spurious regression** Granger and Newbold (1974): OLS of one $I(1)$ series on another, unrelated one

$$y_t = a + b x_t + u_t, \quad x_t, y_t \text{ independent random walks}$$

- ▶ u_t is itself $I(1)$: the usual t and F tests are invalid, $|t| > 1.96$ far more often than 5%
- ▶ typical signs: a high R^2 and strongly autocorrelated residuals
- ▣ **Correlation in levels** of $I(1)$ series proves nothing
 - ▶ compare the correlation of the changes $\Delta x_t, \Delta y_t$, which are stationary
 - ▶ the exception: cointegrated series, for which u_t is stationary (next slides)

Prerequisites for Today (4/15): independent random walks and the t test

- Left: two independent random walks, $T = 200$; they seem to move together, then apart
- Right: the t statistic of $\hat{b}$ in 2000 such regressions, against the $N(0, 1)$ reference
 - ▶ $|t| > 1.96$ in 82% of the pairs instead of 5%; median R^2 : 0.17, with no relation at all

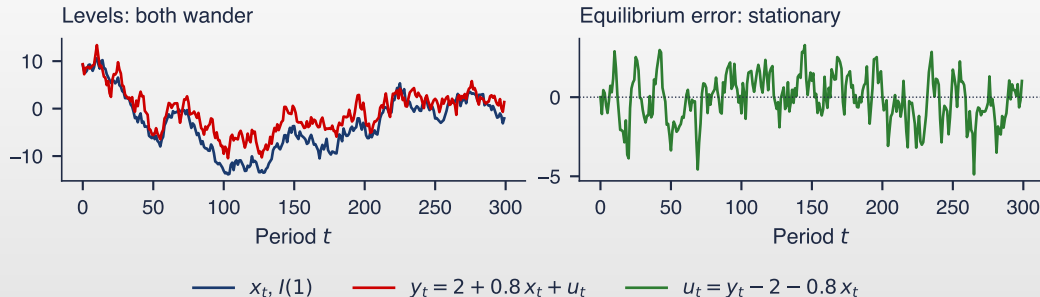
Prerequisites for Today (5/15): cointegration

- **Cointegration** Engle and Granger (1987): $I(1)$ series with a stationary linear combination

$$\mathbf{y}_t = (y_{1t}, \dots, y_{nt})^\top \sim I(1), \quad u_t = \beta^\top \mathbf{y}_t - \mu \sim I(0)$$

- ▶ $\mathbf{y}_t$: the vector of the n series; $^\top$: the transpose; β : the cointegrating vector, normalised with one coefficient equal to 1
- ▶ u_t : the equilibrium error, μ : its mean; two series: $y_{1t} = \mu + b y_{2t} + u_t$, $\beta = (1, -b)^\top$
- **Common trends** Stock and Watson (1988): n variables, r cointegrating vectors, $n - r$ common stochastic trends
 - ▶ example: the 1-year and the 10-year interest rate wander, but their spread stays within a band
 - ▶ a known β , such as $(1, -1)$ for $c_t - y_t$ (log consumption minus log GDP, a “great ratio”): test ADF directly on $\beta^\top \mathbf{y}_t$

Prerequisites for Today (6/15): a cointegrated pair



- Simulated: x_t a random walk, $y_t = 2 + 0.8x_t + u_t$ with $u_t = 0.7u_{t-1} + e_t$ stationary
- Left: both levels wander between about -14 and 11 , together
 - ▶ right: the equilibrium error u_t returns to 0 (standard deviation 1.5); $\beta = (1, -0.8)^\top$, one common trend

Prerequisites for Today (7/15): the Engle–Granger test

- ▣ **Step 1:** OLS in levels, then the residuals

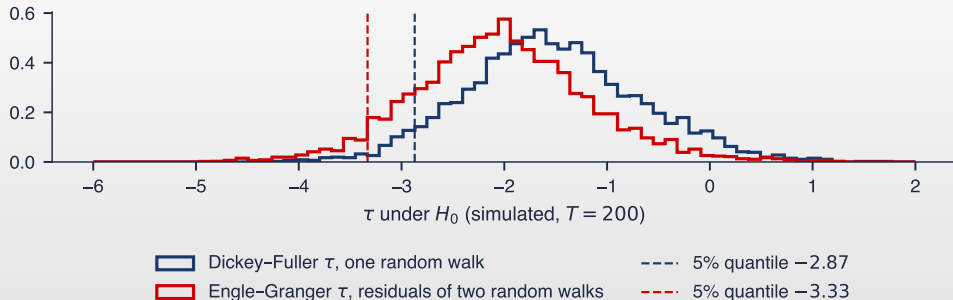
$$y_t = a + b x_t + u_t, \quad \hat{u}_t = y_t - \hat{a} - \hat{b} x_t$$

- ▣ **Step 2:** an ADF test on $\hat{u}_t$, without constant ($\hat{u}_t$ has mean 0)

$$\Delta \hat{u}_t = \gamma \hat{u}_{t-1} + \sum_{j=1}^k \delta_j \Delta \hat{u}_{t-j} + e_t, \quad \tau = \frac{\hat{\gamma}}{\text{SE}(\hat{\gamma})}$$

- ▶ H_0 : no cointegration ($\gamma = 0$, $\hat{u}_t$ has a unit root); reject if τ is below the critical value
- ▣ **Critical values** MacKinnon (1994), constant, large T , 5%: more negative than Dickey–Fuller
 - ▶ -2.86 (one series, Dickey–Fuller); -3.34 (2 variables); -3.74 (3 variables)
 - ▶ **Phillips–Ouliaris** Z_t Phillips and Ouliaris (1990): the same H_0 and critical values; no lags, a long-run variance correction instead

Prerequisites for Today (8/15): why the critical values are more negative



- Simulated τ under H_0 , $T = 200$, 4000 replications: blue for one random walk (constant), red for the residuals of two independent random walks
- 5% quantiles: -2.87 and -3.33 , close to -2.86 and -3.34
 - ▶ OLS chooses $\hat{b}$ to minimise the variance of $\hat{u}_t$, so $\hat{u}_t$ looks more stationary than it is

Prerequisites for Today (9/15): the error correction model

- **ECM** (error correction model) Davidson et al. (1978): the change of y reacts to last period's distance from equilibrium

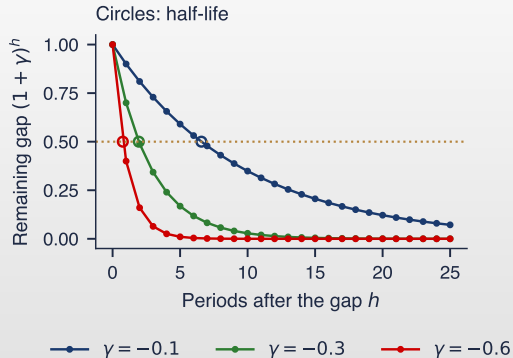
$$\Delta y_t = c + \gamma \hat{u}_{t-1} + \delta_0 \Delta x_t + \dots + \varepsilon_t, \quad \hat{u}_{t-1} = y_{t-1} - \hat{a} - \hat{b} x_{t-1}$$

- ▶ $\gamma < 0$: the **speed of adjustment**, the share of last period's gap closed by y ; $\gamma \geq 0$: no error correction
- ▶ δ_0 : the short-run effect of x ; b : the long-run effect; c : the constant; "...": lags of Δy and Δx ; ε_t : white noise
- ▶ all the terms are stationary: OLS and t -tests are valid
- **Half-life**: without new shocks a gap shrinks like $(1 + \gamma)^h$

$$(1 + \gamma)^{h_{1/2}} = 0.5 \iff h_{1/2} = \frac{\ln 0.5}{\ln(1 + \gamma)}$$

- ▶ example: $\gamma = -0.5$ gives $h_{1/2} = 1$; $\gamma = -0.1$ gives about 6.6 periods

Prerequisites for Today (10/15): how fast a gap closes



- A gap of 1 at $h = 0$
 - ▶ remaining after h periods: $(1 + \gamma)^h$
- Half-lives (circles)
 - ▶ $\gamma = -0.1$: 6.6 periods
 - ▶ $\gamma = -0.3$: 1.9 periods
 - ▶ $\gamma = -0.6$: 0.8 periods
- With monthly data, $\gamma = -0.1$
 - ▶ means half a year to close half of a gap

Prerequisites for Today (11/15): the VECM

- ▣ **VECM** (vector error correction model): a VAR(p) of [Chapter 6](#) rewritten in differences

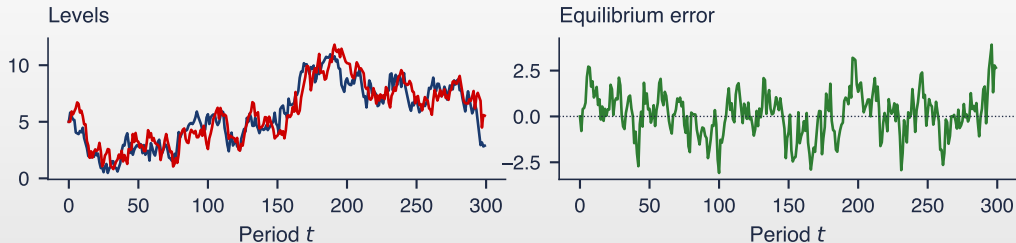
$$\Delta \mathbf{y}_t = \mathbf{\Pi} \mathbf{y}_{t-1} + \sum_{i=1}^{p-1} \mathbf{\Gamma}_i \Delta \mathbf{y}_{t-i} + \mathbf{u}_t, \quad \mathbf{\Pi} = \mathbf{A}_1 + \cdots + \mathbf{A}_p - \mathbf{I}, \quad \mathbf{\Gamma}_i = -(\mathbf{A}_{i+1} + \cdots + \mathbf{A}_p)$$

- ▶ from $\mathbf{y}_t = \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t$; $\mathbf{A}_i$: the $n \times n$ VAR matrices; $\mathbf{I}$: the identity; $\mathbf{u}_t$: white-noise shocks
- ▣ $r = \text{rank}(\mathbf{\Pi})$: the number of cointegrating vectors

$$0 < r < n: \quad \mathbf{\Pi} = \boldsymbol{\alpha} \boldsymbol{\beta}^\top, \quad \boldsymbol{\alpha}, \boldsymbol{\beta} \text{ of size } n \times r$$

- ▶ $r = 0$: no cointegration (a VAR in differences); $r = n$: all series stationary
- ▶ columns of $\boldsymbol{\beta}$: the cointegrating vectors; $\boldsymbol{\alpha}$: the **adjustment coefficients** (loadings) of each variable to each equilibrium error
- ▶ $\alpha_i = 0$: y_i does not adjust, it is **weakly exogenous**
- ▶ **Granger representation theorem** [Engle and Granger \(1987\)](#): with cointegration at least one $\alpha_i \neq 0$

Prerequisites for Today (12/15): a VECM with one adjusting variable



— $y_{2t}, \alpha_2 = 0$ (does not adjust)
 — $y_{1t}, \alpha_1 = -0.2$ (adjusts)
 — $z_t = \boldsymbol{\beta}^\top \mathbf{y}_t = y_{1t} - y_{2t}$

- $\Delta y_{1t} = \alpha_1 z_{t-1} + u_{1t}, \Delta y_{2t} = \alpha_2 z_{t-1} + u_{2t}$, with $z_t = y_{1t} - y_{2t}$, $\boldsymbol{\beta} = (1, -1)^\top$
- $\alpha_1 = -0.2, \alpha_2 = 0$: y_2 is a random walk (weakly exogenous), y_1 follows it
 - ▶ the levels share one common trend, the trend of y_2 ; z_t stays around 0

Prerequisites for Today (13/15): the Johansen tests

- ▣ **Johansen Johansen (1988)**: maximum likelihood for the VECM gives eigenvalues $1 > \hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_n \geq 0$
 - ▶ $\hat{\lambda}_i$: the squared canonical correlation between $\Delta \mathbf{y}_t$ and a combination of $\mathbf{y}_{t-1}$; large: a stationary combination; ≈ 0 : still $I(1)$
- ▣ Two statistics, with T the number of observations

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_i), \quad \lambda_{\text{max}}(r) = -T \ln(1 - \hat{\lambda}_{r+1})$$

- ▶ trace: H_0 : rank $\leq r$ against rank $= n$; maximum eigenvalue: H_0 : rank $= r$ against rank $= r + 1$
- ▶ large statistics (eigenvalues far from 0): reject H_0 ; non-standard critical values **Osterwald-Lenum (1992)**
- ▣ **Sequential procedure**: test $r = 0, r \leq 1, \dots$; the rank is the first r not rejected
 - ▶ 5% critical values with a constant, $n = 3$: trace 29.80, 15.49, 3.84; max 21.13, 14.26, 3.84
 - ▶ **restricted constant**: the constant enters only $\beta^\top \mathbf{y}_{t-1} + \mu$, the levels have no drift

Prerequisites for Today (14/15): pairs trading

- **Spread** of two cointegrated prices [Gatev, Goetzmann and Rouwenhorst \(2006\)](#), standardised with formation-period moments

$$s_t = \ln P_t^A - \hat{b} \ln P_t^B, \quad z_t = \frac{s_t - \bar{s}}{\hat{\sigma}_s}$$

- ▶ P_t^A, P_t^B : the two prices; $\hat{b}$: the Engle–Granger slope; $\bar{s}, \hat{\sigma}_s$: mean and standard deviation of s_t in the formation period
- **Rule**: open at $|z_t| > 2$, close when z_t returns to 0
 - ▶ $z_t > 2$: sell A, buy $\hat{b}$ units of B (short the spread); $z_t < -2$: the reverse (long the spread)
 - ▶ parameters fixed in the formation period, applied in a later trading period: an out-of-sample test
- **Sharpe ratio**: return per unit of risk
$$\text{SR} = \frac{\bar{r}_{\text{annual}}}{\sigma_{\text{annual}}}$$
 - ▶ $\bar{r}_{\text{annual}}$: the mean annual return of the strategy; σ_{annual} : its annual standard deviation; net: after transaction costs

Prerequisites for Today (15/15): the z-score rule



- A simulated stationary spread ($s_t = 0.95 s_{t-1} + e_t$), 500 days, standardised
- 4 trades: each opens at ± 2 (dashed) and closes at 0 (dotted); 25% of the days in the market
 - ▶ the rule earns only if the spread keeps returning to 0, that is, if the cointegration survives the formation period

A1: an Engle–Granger statistic [Solved]

Task

- ▣ Two $I(1)$ series, $T = 200$. Step 1 gives residuals $\hat{u}_t$; step 2 gives $\Delta\hat{u}_t = -0.118\hat{u}_{t-1}$, with $SE = 0.036$.
- ▣ 1. Compute τ .
- ▣ 2. Compare τ with the 5% and 10% Engle–Granger values for 2 variables, -3.37 and -3.07 , and decide.
- ▣ 3. Say what the Dickey–Fuller value -2.88 would have concluded.
- ▣ 4. Explain why the two critical values differ.
- ▣ Report: one number, two decisions and one sentence.

Solution

- ▣ 1. $\tau = -0.118/0.036 = -3.28$
- ▣ 2. $-3.28 > -3.37$: do not reject “no cointegration” at 5% ($p = 0.058$); $-3.28 < -3.07$: reject at 10%
- ▣ 3. $-3.28 < -2.88$: the Dickey–Fuller table would wrongly report cointegration at 5%
- ▣ 4. OLS chooses $\hat{b}$ to minimise the residual variance, so $\hat{u}_t$ looks more stationary than a true error: the null distribution shifts to the left.

A2: three variables [Proposed]

Solution discussed in class

Task

- ▣ Three $I(1)$ series, $T = 150$: y_t regressed on x_{1t} and x_{2t} . Step 2 gives $\hat{\gamma} = -0.162$, $SE = 0.041$. Model: A1.
- ▣ 1. Compute τ .
- ▣ 2. Decide at 5% and at 1% with the values for 3 variables, -3.80 and -4.39 .
- ▣ 3. Say which critical value a student would wrongly use if he forgot x_{2t} (2 variables: -3.38).
- ▣ 4. Say how many cointegrating vectors the Engle–Granger method can find with three variables.
- ▣ Report: one number, two decisions and two sentences.

A3: an error correction model [Solved]

Task

- ▣ Estimated ECM (log consumption c_t , log income y_t , in %): $\Delta c_t = 0.2 + 0.5 \Delta y_t - 0.25 (c_{t-1} - 0.9 y_{t-1})$.
- ▣ 1. Give the long-run and the short-run effect of income on consumption.
- ▣ 2. Interpret the coefficient -0.25 and compute the half-life.
- ▣ 3. Compute Δc_t if $c_{t-1} - 0.9 y_{t-1} = 2$ and $\Delta y_t = 1$.
- ▣ 4. Say what a coefficient of $+0.25$ would mean.
- ▣ Report: two elasticities, one half-life, one forecast and one sentence.

Solution

- ▣ 1. Long run: 0.9 (from the equilibrium $c = 0.9y + \text{const.}$); short run: 0.5 (same period)
- ▣ 2. A quarter of last period's gap is closed each period;
 $h_{1/2} = \ln 0.5 / \ln 0.75 = 2.41$ periods
- ▣ 3. $\Delta c_t = 0.2 + 0.5 \cdot 1 - 0.25 \cdot 2 = 0.2$: the error correction cancels the short-run effect
- ▣ 4. Consumption would move away from equilibrium after a gap: no error correction, no cointegration.

A4: a bivariate VECM [Proposed]

Solution discussed in class

Task

- $\Delta y_{1t} = -0.20 z_{t-1} + u_{1t}$, $\Delta y_{2t} = 0.05 z_{t-1} + u_{2t}$, with $z_t = y_{1t} - y_{2t}$. Model: A3.
- 1. Write α , β and $\Pi = \alpha\beta^\top$, and give the rank of Π .
- 2. Explain which variable does most of the adjusting.
- 3. Show that $z_t = (1 + \beta^\top \alpha) z_{t-1} + (u_{1t} - u_{2t})$ and compute its half-life.
- Report: one matrix, one rank, one half-life and one sentence.

A5: Johansen statistics from eigenvalues [Solved]

Task

- ▣ Three $I(1)$ series, $T = 200$, case with a constant; the Johansen eigenvalues are 0.120, 0.045 and 0.008.
- ▣ 1. Compute the trace statistics for $r = 0$, $r \leq 1$ and $r \leq 2$.
- ▣ 2. Compute the maximum-eigenvalue statistics.
- ▣ 3. Compare with the 5% values (trace 29.80, 15.49, 3.84; maximum 21.13, 14.26, 3.84) and choose the rank.
- ▣ 4. Give the number of common trends.
- ▣ Report: six statistics, one rank and one number of trends.

Solution

- ▣ 1. $-200[\ln 0.880 + \ln 0.955 + \ln 0.992] = 36.38$;
 $-200[\ln 0.955 + \ln 0.992] = 10.82$; $-200 \ln 0.992 = 1.61$
- ▣ 2. 25.57; 9.21; 1.61
- ▣ 3. $r = 0$ rejected by both ($36.38 > 29.80$, $25.57 > 21.13$);
 $r \leq 1$ not rejected: rank 1
- ▣ 4. $n - r = 3 - 1 = 2$ common stochastic trends.

A6: from a VAR(2) to a VECM [Proposed]

Solution discussed in class

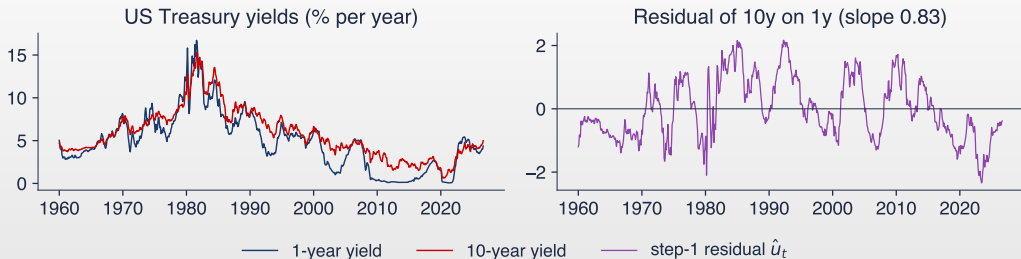
Task

- ▣ $\mathbf{y}_t = \mathbf{A}_1 \mathbf{y}_{t-1} + \mathbf{A}_2 \mathbf{y}_{t-2} + \mathbf{u}_t$ with $\mathbf{A}_1 = \begin{pmatrix} 0.7 & 0.2 \\ 0.1 & 0.7 \end{pmatrix}$, $\mathbf{A}_2 = \begin{pmatrix} 0.1 & 0 \\ 0 & 0.2 \end{pmatrix}$. Model: A5.
- ▣ 1. Compute $\Pi = \mathbf{A}_1 + \mathbf{A}_2 - \mathbf{I}$ and $\Gamma_1 = -\mathbf{A}_2$.
- ▣ 2. Show that Π has rank 1 and write it as $\alpha\beta^\top$ with $\beta = (1, -1)^\top$.
- ▣ 3. Interpret the signs of α .
- ▣ Report: two matrices, α and one sentence.

B1: the 1-year and the 10-year US yields [Solved]

- **Question:** are the 1-year and the 10-year Treasury yields cointegrated, how fast does their gap close, and which rate closes it?
- **Inputs:** FRED GS1 and GS10, monthly averages, % per year, January 1960–September 2026
- **Tasks**
 1. Run ADF (with a constant) on both yields and on the change of the 10-year yield.
 2. Run Engle–Granger (10-year on 1-year) and Phillips–Ouliaris, and compare with the 5% Engle–Granger value.
 3. Estimate the ECM of the 10-year yield (one lag of each difference) and the same equation for the 1-year yield.
 4. Interpretation: if the 10-year yield is 1 percentage point above its equilibrium, how long until half of the gap is closed, and by which rate?
- **Report:** three ADF p-values, two cointegration tests, two adjustment coefficients and one sentence
- **Notebook:** section B1

B1: solution [Solved]



- ▣ ADF $p = 0.26$ (1 year), 0.63 (10 years): unit roots; the change: $p < 0.001$; both yields $I(1)$
- ▣ Step 1: $\hat{v}_t^{(10)} = 1.76 + 0.827 i_t^{(1)}$; step 2: $\tau = -3.59$ (19 lags) < -3.34 : cointegration at 5% ($p = 0.025$); Phillips–Ouliaris $Z_t = -3.65$ ($p = 0.022$)
- ▣ ECM, 10 years: $\hat{\gamma} = -0.0295$ ($t = -4.81$), $\hat{\delta}_0 = 0.53$; 1 year: $\hat{\gamma} = 0.0172$ ($t = 1.25$), not significant
- ▣ Interpretation: half-life $\ln 0.5 / \ln(1 - 0.0295) = 23.1$ months, about two years; the 10-year yield closes the gap, the 1-year yield (set by monetary policy) does not react

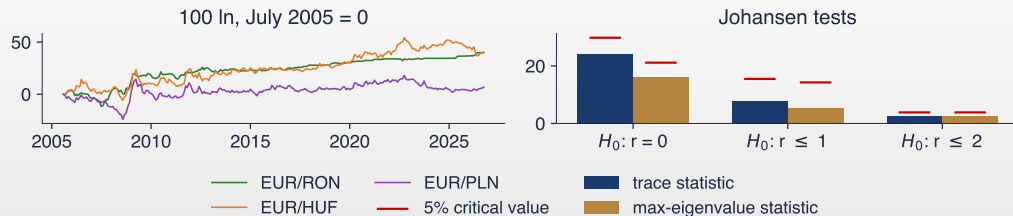
B2: consumption and GDP in Romania [Proposed]

- **Question:** is Romanian household consumption cointegrated with GDP, as the “great ratios” suggest?
- **Inputs:** Eurostat, real household consumption and real GDP, chain-linked volumes, seasonally adjusted, 1995–2026; c_t , y_t as 100 ln; model: B1
- **Tasks**
 1. Run ADF (constant and trend) on c_t and y_t , and ADF (constant) on their changes.
 2. Run Engle–Granger (c_t on y_t) and Phillips–Ouliaris, and report the slope.
 3. Plot the ratio of consumption to GDP and run ADF on $c_t - y_t$.
 4. Interpretation: does a stable consumption share make sense for Romania over 1995–2026?
- **Report:** four ADF tests, two cointegration tests, one chart and two sentences
- **Notebook:** section B2

B3: three Central European currencies [Solved]

- ▣ **Question:** do EUR/RON, EUR/HUF and EUR/PLN share a long-run equilibrium?
- ▣ **Inputs:** BNR reference rates (cross rates for HUF and PLN), month-end, 100 ln, since July 2005
- ▣ **Tasks**
 1. Run ADF on each rate.
 2. Run the Johansen trace and maximum-eigenvalue tests with a constant and lags by BIC.
 3. Compare with the three pairwise Engle–Granger tests and with the correlations of levels and of changes.
 4. Interpretation: should the three rates be modelled with a VECM?
- ▣ **Report:** three ADF p-values, a table of six Johansen statistics, a rank and one sentence
- ▣ **Notebook:** section B3

B3: solution [Solved]



	$r = 0$	$r \leq 1$	$r \leq 2$
trace (5% value)	24.21 (29.80)	7.94 (15.49)	2.72 (3.84)
maximum eigenvalue (5% value)	16.27 (21.13)	5.21 (14.26)	2.72 (3.84)

- ADF $p = 0.55; 0.75; 0.08$: three $I(1)$ rates; Johansen ($T = 255, k = 1$): no statistic exceeds its critical value: rank 0
- Engle–Granger $p = 0.29$ (RON, HUF), 0.08 (RON, PLN), 0.20 (HUF, PLN); correlation 0.89 in levels but only 0.31 in monthly changes
- Interpretation: no; with rank 0 there is no β to estimate; model the changes with a VAR in differences (Chapter 6)

B4: ROBOR, the Romanian 10-year yield and Euribor [Proposed]

- **Question:** how many long-run relations tie Romanian interest rates to each other and to Euribor, and which rate adjusts?
- **Inputs:** ROBOR 3M and Euribor 3M (Eurostat), Romanian 10-year yield (monthly mean of daily data), January 2010–August 2026; model: B3
- **Tasks**
 1. Run the Johansen tests (constant, BIC lags) and choose the rank.
 2. Estimate the VECM with that rank and the constant restricted to the cointegrating relation; normalise β on ROBOR.
 3. Report $\hat{\alpha}$ with t -statistics and run ADF on the equilibrium error.
 4. Interpretation: which rates are weakly exogenous, and what does this say about Romanian monetary conditions?
- **Report:** a table of Johansen statistics, $\hat{\beta}$, $\hat{\alpha}$ with t , one ADF test and two sentences
- **Notebook:** section B4

C1: pairs trading on two banks [Proposed]

- **Question:** would a trader who found Banca Transilvania and BRD cointegrated at the end of 2021 have earned money afterwards?
- **Inputs:** adjusted daily prices, formation 2014–2021, trading January 2022–September 2026; models: B1, B3
- **Tasks**
 1. On the formation period, run Engle–Granger and Phillips–Ouliaris (TLV on BRD) and keep $\hat{b}$, the mean and the standard deviation of the spread.
 2. On the trading period, apply the rule: open at $|z| > 2$, close at $z = 0$ (z : the spread minus its formation mean, divided by its formation standard deviation); compute the annual return and the Sharpe ratio (mean annual return divided by its annual standard deviation), gross and net of 0.20% per unit traded.
 3. Repeat the Engle–Granger test on the trading period alone.
 4. Interpretation: is the result evidence that pairs trading works on the BVB?
- **Report:** two tests, the number of trades, two returns, two Sharpe ratios and a plan for a project
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant about cointegration. The answer:
- (a) The Engle-Granger statistic is -3.0 with two variables; since $-3.0 < -2.87$, the 5% Dickey-Fuller value, the series are cointegrated.
- (b) The ECM coefficient of the lagged equilibrium error is +0.15, so 15% of the gap is corrected each period.
- (c) Johansen rejects $r = 0$ and $r \leq 1$ but not $r \leq 2$ for three yields, so the yields share two common trends.
- (d) EUR/RON and EUR/HUF are both $I(1)$ and $R^2 = 0.79$ in levels, so they are cointegrated.
- (e) If two series are cointegrated, at least one of them Granger-causes the other.
- (f) A pairs-trading rule with a Sharpe ratio of 0.44 on 2014-2026 data will keep this Sharpe ratio in the future.
- **Tasks**
 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
- **Report:** a list of six verdicts with one line of justification each

Key takeaways

- ▣ Test that every series is $I(1)$ first; then test the equilibrium error, never the correlation of levels
- ▣ Residual-based tests need Engle–Granger critical values, which depend on the number of variables
- ▣ The ECM gives the speed of adjustment; the half-life $\ln 0.5 / \ln(1 + \gamma)$ is in the units of the data
- ▣ Johansen gives the rank; $n - r$ is the number of common trends; α shows who adjusts
- ▣ An AI answer is a draft: check the critical values, the signs and the number of trends

After the seminar

- Lecture 7 develops each topic of today
 - ▶ common trends, the Engle–Granger distribution, ECM, the VECM and the Granger representation theorem, the Johansen tests, forecasting and pairs trading
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: pairs trading on the BVB, out of sample and after costs
- Reading: [Huang and Petukhina \(2022\)](#), Ch. 9; [Hamilton \(1994\)](#), Ch. 19–20; [Engle and Granger \(1987\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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